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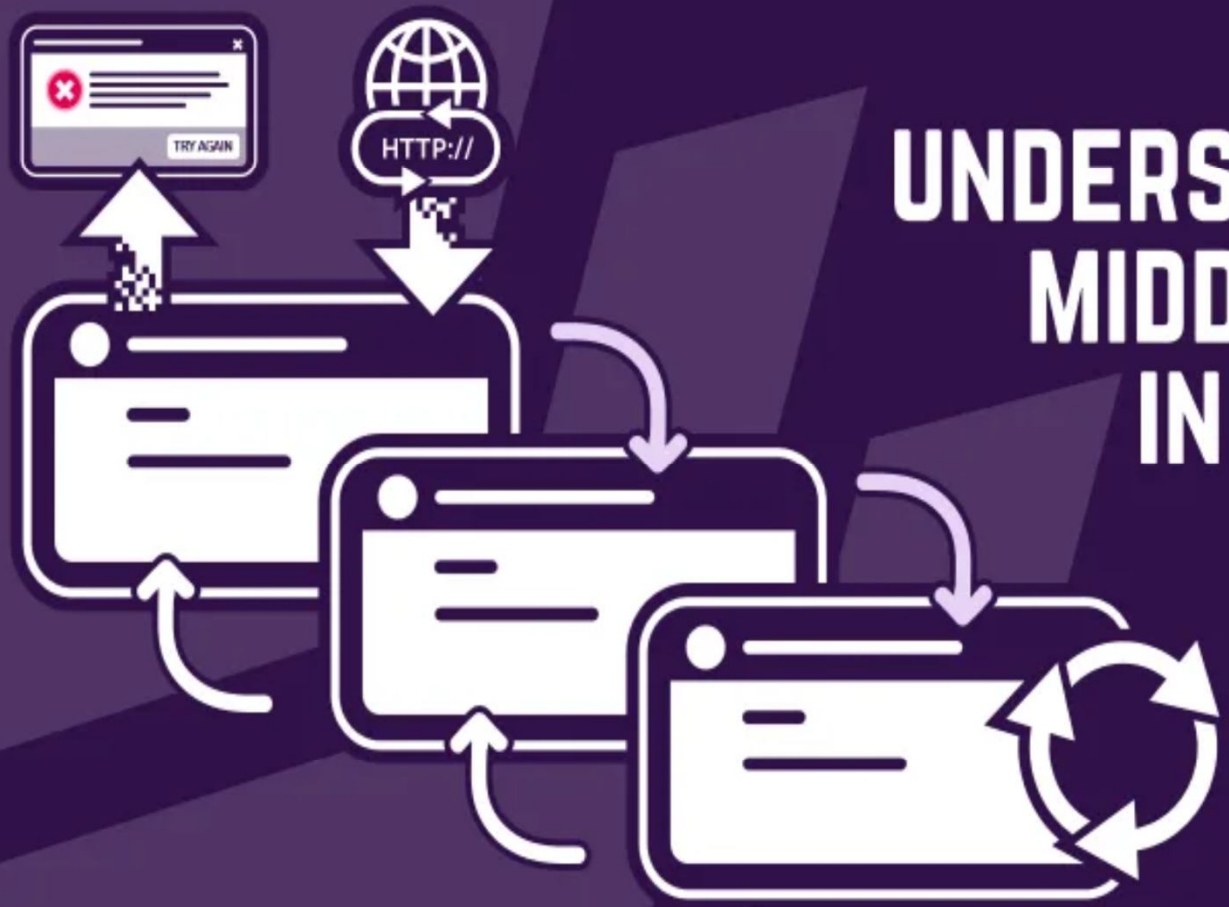

UNDERSTANDING MIDDLEWARE IN ASP.NET CORE

By Achyut V. Kendre

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Today's Learning.

- Startup Class Flash Back.
- Configure Service and Configure Method.
- What is Request Pipeline?
- What is Middleware?
- How Middleware Works?
- Run & Use Method?

Startup Class Flash Back

- The ASP.NET Core 3.1, and ASP.NET Core 5 Applications must include a Startup Class.
- It is like the Global.asax file our traditional .NET Application.
- As the name suggests, it is executed first when the application starts.
- The Startup class can be configured using UseStartup<T>() Extension method at the time of Configuring the Host in the Main() Method of the Program class.

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Startup Class Flash Back

- The name “Startup” is by the ASP.NET Core Convention.
- However, you can give any name to the Startup class, just specify it as the generic parameter in the UseStartup<T>() method.
- Startup class must includes two public methods: ConfigureServices and Configure. The Startup class must include the Configure method and can optionally include the ConfigureService method.

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Configure Services Method

- ASP.NET Core built from scratch to Support Dependency Injections.
- It includes the built-in IoC container to provide dependent objects using constructors.
- The ConfigureServices method is a place where you can register your dependent classes with the built-in IoC container.
- After registering the dependent classes, it can be used anywhere in the application.

Configure Services Method

- constructor of a class where you want to use it. You just need to include it in the parameter of the
- The IoC container will inject it automatically.
- ASP.NET Core Refers to the dependent class as a Service. So, whenever you read “Service” then understand it as a class that is going to be used in some other class.
- The ConfigureServices method includes the IServiceCollection parameter to register services to the IoC container.

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Configure( ) Method

- The Configure method is a place where we can configure the application request pipeline for our ASP.NET Core application using the IApplicationBuilder instance that is provided by the built-in IoC container.
- ASP.NET Core introduced the middleware components to define a request pipeline, which will be executed on every request.

Configure( ) Method

- You need to include only those middleware components which are required by your application and thus increase the performance of your application.

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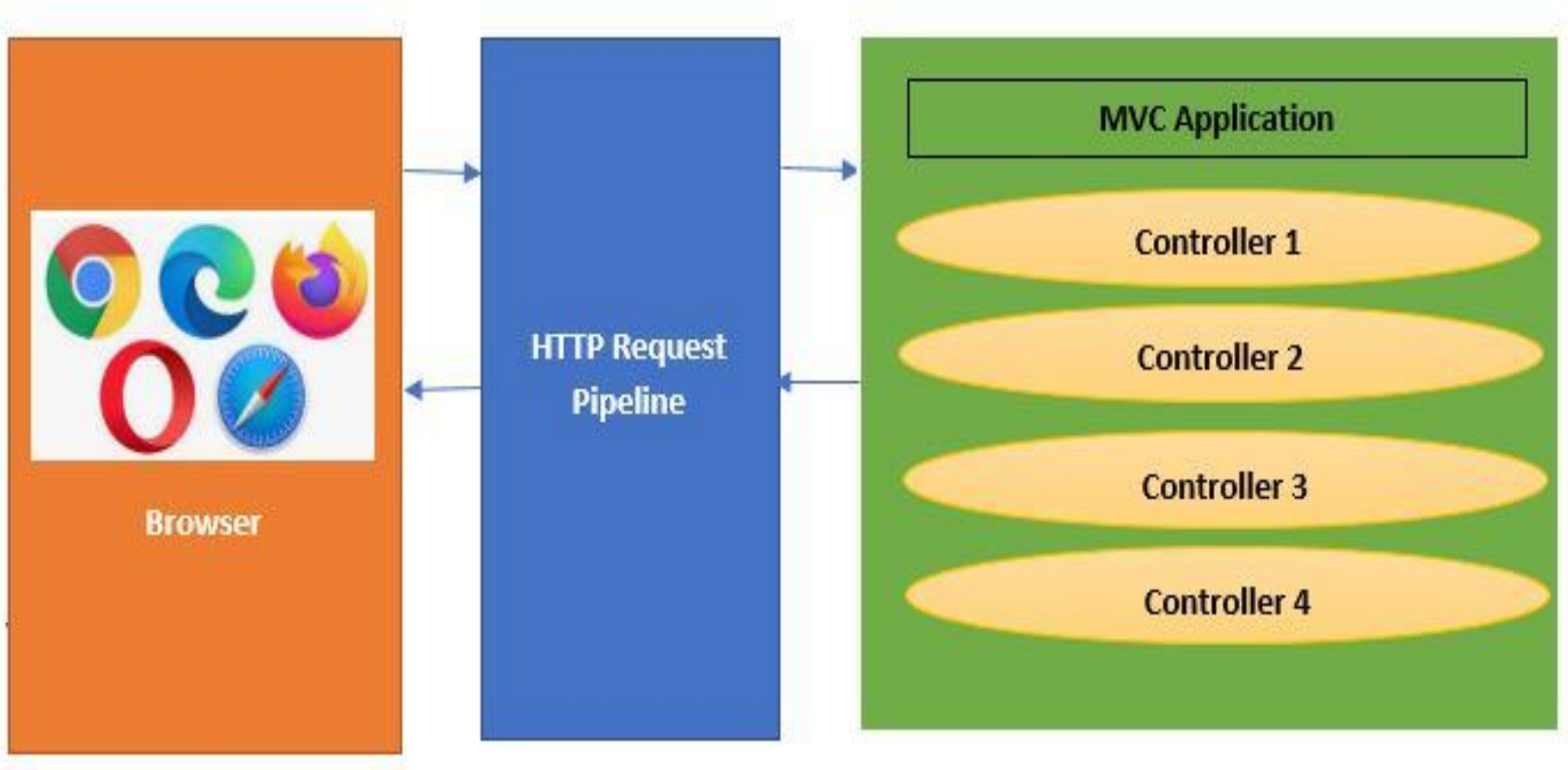
Request Pipe Line



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Request Pipeline

As per the above diagram -

- The browser sent a request to the HTTP request pipeline.
- HTTP pipeline will execute.
- Once HTTP pipeline execution will be completed, the request goes to the MVC application.
- MVC Application executes the request and sends the response back to the HTTP Request pipeline -> Browser.

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What is Middleware?

- Middleware is new Concept in ASP.NET Core.
- ASP.NET Core Middleware is a fundamental concept in ASP.NET Core applications.
- Middleware can be a function/class/functional unit will perform some specific task.
- Middleware components are the building blocks of the request processing pipeline in an ASP.NET Core application.

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What is Middleware?

- They handle specific tasks during request processing, such as authentication, routing, logging, and more.
- Middleware components allow you to add functionalities in a modular and customizable manner.
- Middleware acts as a bridge between the web server and your application code.

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What is Middleware?

- It processes HTTP requests and responses.
- Middleware components are arranged in a pipeline, where each component performs specific tasks.
- The pipeline starts with the web server and ends with your application code.

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How middleware's work?

- Has Access to both Request & Response
- May simply pass the Request to next Middleware
- May process and then pass the Request to next Middleware
- May handle the request and short circuit the pipeline
- May Process the outgoing Response.
- Middleware's are executed in the order they area added.

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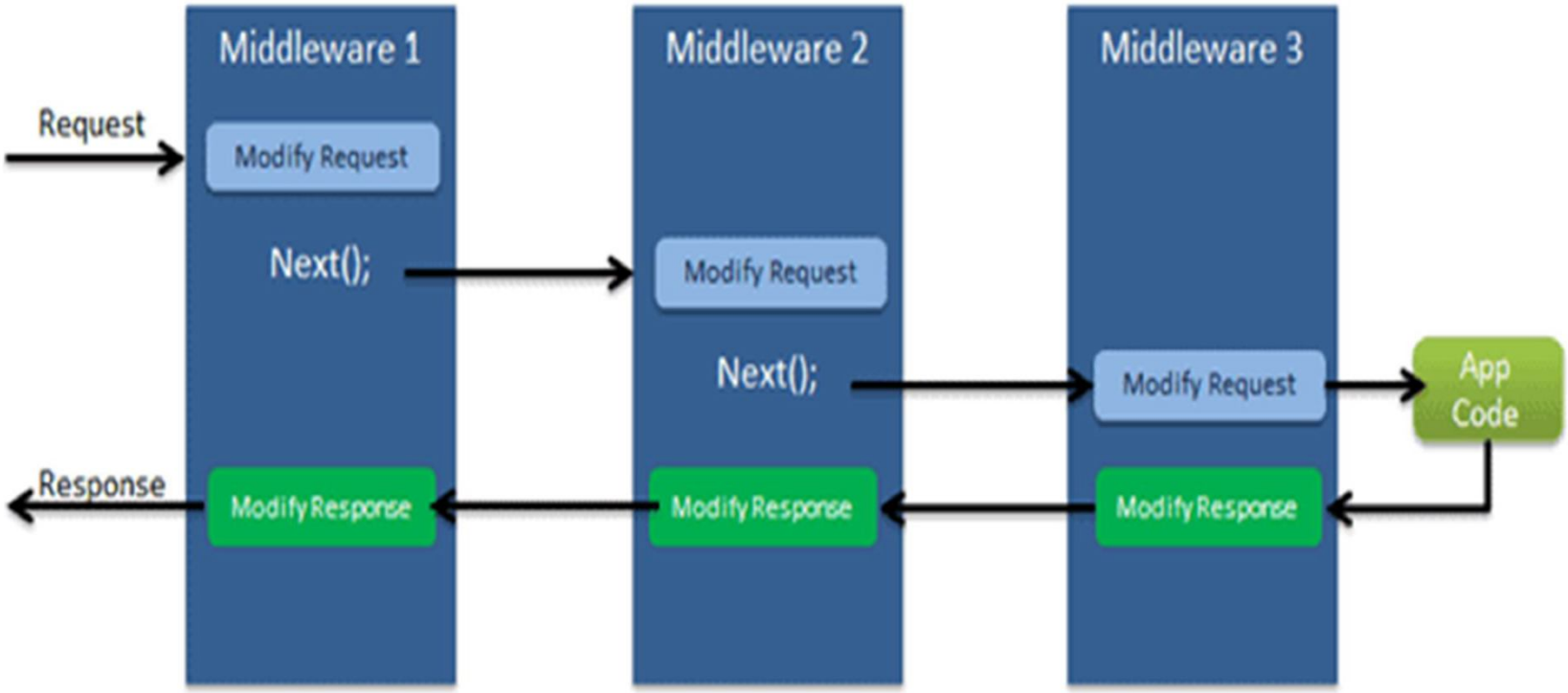
What is Middleware?



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What is Middleware?

- Configure middleware in the Configure method of the Startup class using IApplicationBuilder instance.
- Middleware's can be built in or can be user defined.
- IApplicationBuilder provides run and use method to create user defined middleware.

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Run Method

- The Run method is an extension method on IApplicationBuilder & accepts a RequestDelegate as parameter.
- The RequestDelegate is a delegate method which handles the request.
- RequestDelegate stores a methods who has HttpContext object as parameter.
- Run method has limitation that it will short circuit the Reqeust Pipe Line.

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Run Method

```
app.Run(async(context)=>{
 await context.Response.WriteAsync("Hello World!");
});
```

```
Public void Configure(IApplicationBuilder app,
IHostingEnvironment env) {
app.Run(MyMiddleware);
}

private Task MyMiddleware(HttpContext context){
return context.Response.WriteAsync("Hello World! ");
}
```

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Use Method

- Use method is used to configure multiple middleware.
- Use() method has RequestDelegate as parameter who has two parameters.
- The first parameter is the HttpContext & second parameter is the Func type(generic delegate) that represents next middleware in the pipeline.

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Use Method

```
app.Use(async (context, next) => {
await context.Response.WriteAsync("Hello World From
1st Middleware!");
await next(); });
```

THANKS!

Any questions?

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